

CASE STUDY

CO₂-CURED COPPER SLAG CEMENT

Replacing 20% of cement with copper slag and curing with CO₂ made mortar up to **67% stronger at 3 days**, locked carbon into the material and cut net emissions by **35.5% per tonne of binder**, while the slag helped offset the cost of carbonation.



Locking CO₂ Into Concrete While Recycling Mining Waste

The cement industry produces around 8% of global CO₂ emissions. Copper production, meanwhile, left about 37.7 million tonnes of copper slag in 2019 alone, a waste that is costly to handle and store. This work, carried out at the University of Arizona with City University of Hong Kong, combined the two problems into one solution: use copper slag in place of some cement, then cure the young material with CO₂ so it absorbs carbon and gains strength.

OVERVIEW

Carbonation curing exposes freshly made concrete to concentrated CO₂. The gas reacts with the cement to form calcium carbonate, which fills pores, adds strength and stores carbon permanently.

Copper slag alone is only weakly reactive, but it opens up the cement matrix, giving CO₂ more room to penetrate and carbonates more space to form. Together they turn two waste streams into a stronger, lower-carbon binder.

THE PROJECT

Mortars were made with copper slag replacing 0%, 10% and 20% of the cement. After a controlled drying step, specimens were cured in a sealed vessel at over 95% CO₂ for 2, 6, 12, 24 or 48 hours, then cured normally.

The team measured strength at 3 and 28 days, ultrasonic pulse velocity and CO₂ uptake, characterised the products with XRD, TGA, FTIR and SEM, and estimated the cost and carbon balance per tonne of binder.

At a Glance

- **Feedstock:** copper slag replacing 10% to 20% of Portland cement.
- **Process:** CO₂ curing at over 95% concentration for 2 to 48 hours.
- **Goals:** more strength, more CO₂ stored, lower net emissions.
- **Tests:** strength, ultrasonic pulse velocity and CO₂ uptake.
- **Analysis:** XRD, TGA, FTIR, SEM, plus cost and carbon accounting.



Store CO₂

Turn captured CO₂ into solid carbonate inside the material.



Recycle Mining Waste

Give copper slag a use as a partial cement replacement.



Build Early Strength

Gain strength fast, which matters for precast production.

The Challenge

Copper slag shows only modest pozzolanic activity, so on its own it mostly dilutes the cement. Carbonation curing has its own trade-offs: the material must be partly dried first, which can limit hydration, and liquid CO₂ adds cost.

The question was how long to carbonate, how much slag to use, and whether the combination pays off in strength, stored carbon and cost.

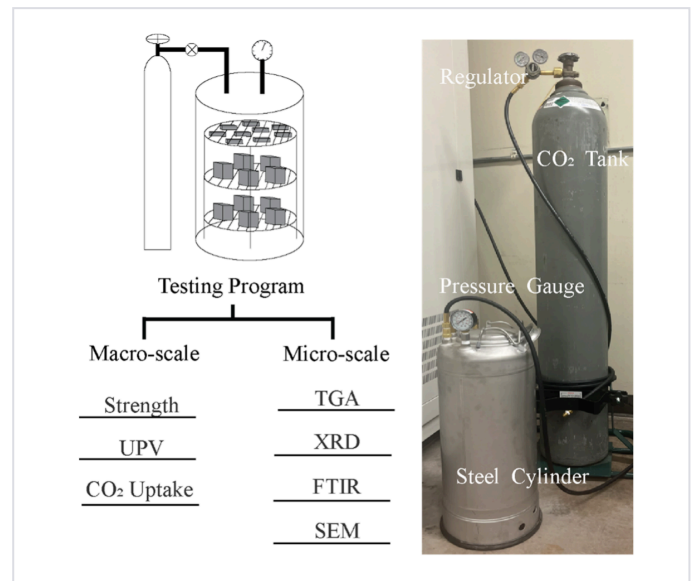
01	Blend Mortars with 0%, 10% and 20% copper slag in place of cement, at a fixed water-to-binder ratio.
02	Pre-condition Specimens were fan-dried until about 35% of their water was removed, opening pores for CO ₂ .
03	Carbonate Cured in a sealed steel vessel at over 95% CO ₂ for 2 hours up to 2 days, then cured normally.
04	Measure and account Strength, pulse velocity, CO ₂ uptake and microstructure, plus cost and emissions per tonne.

What It Means for Manufacturers

- Mining waste and captured CO₂ can both go into the same product.
- Precast plants gain early strength, useful for faster demoulding.
- Slag savings help pay for carbonation, keeping costs close to normal.

How It Works

Pre-conditioned specimens sit on racks inside a steel cylinder fed from a CO₂ tank through a regulator and pressure gauge, followed by a programme of macro- and micro-scale tests.



Results

Strength rose steadily with carbonation time, and 48 hours caused no harm. With 20% slag, 6 to 48 hours of CO₂ curing made mortar 5% to 67% stronger at 3 days and 36% stronger at 28 days. Slag also boosted CO₂ uptake: after 12 hours, 22.6% for the 20% slag mix against 16.7% for plain cement. Very short curing (2 hours) could weaken samples, as drying outweighed the benefit.

Early strength gain

+67%

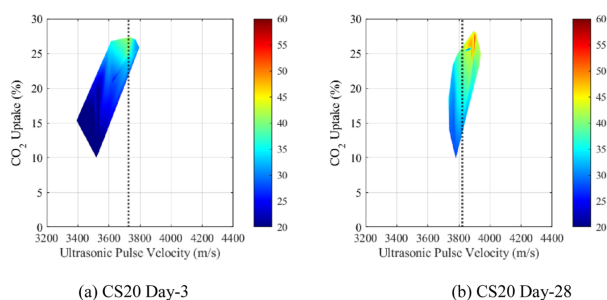
20% slag, 48 h CO₂, day 3

Net CO₂ reduction

35.5%

per tonne of binder

NON-DESTRUCTIVE CHECK · 20% SLAG



Strength Without Breaking Samples

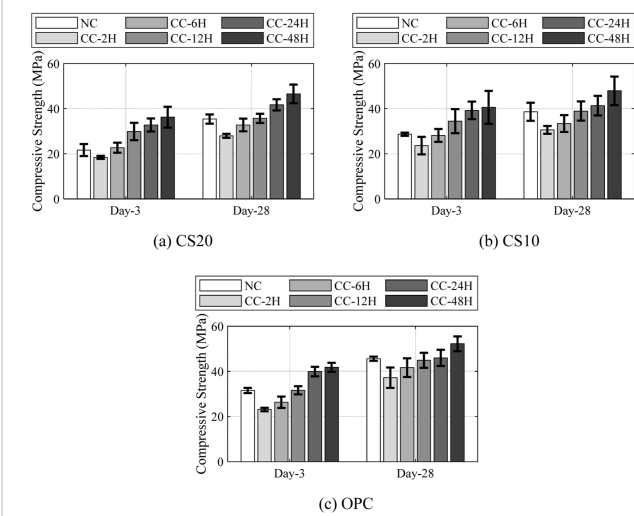
Ultrasonic pulse velocity (x-axis) rises with CO₂ uptake (y-axis) and strength (colour), so a quick, non-destructive pulse test can track how well each batch has carbonated.

Slag mixes ran 3,400 to 3,800 m/s against 3,600 to 4,100 m/s for plain cement.

Results in Detail

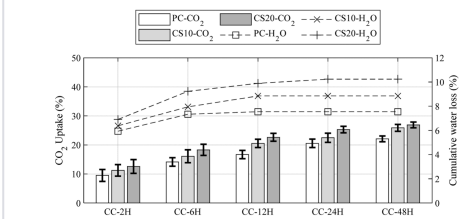
Longer CO₂ curing means more strength and more stored carbon, copper slag amplifies both, and the cost and carbon balance favours the slag blends.

FIGURE 1 · COMPRESSIVE STRENGTH



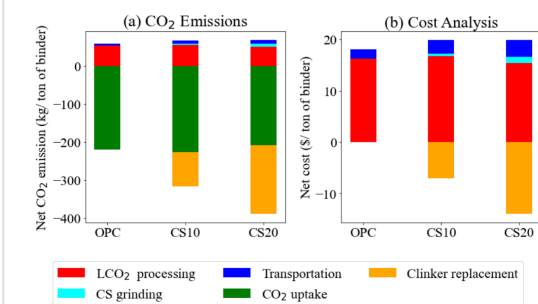
Longer curing, stronger mortar. Strength at day 3 and day 28 for 20% slag (a), 10% slag (b) and plain cement (c), from normal curing (NC) through 2 to 48 hours of CO₂ curing (CC). The slag mixes gained the most.

FIGURE 2 · CO₂ UPTAKE



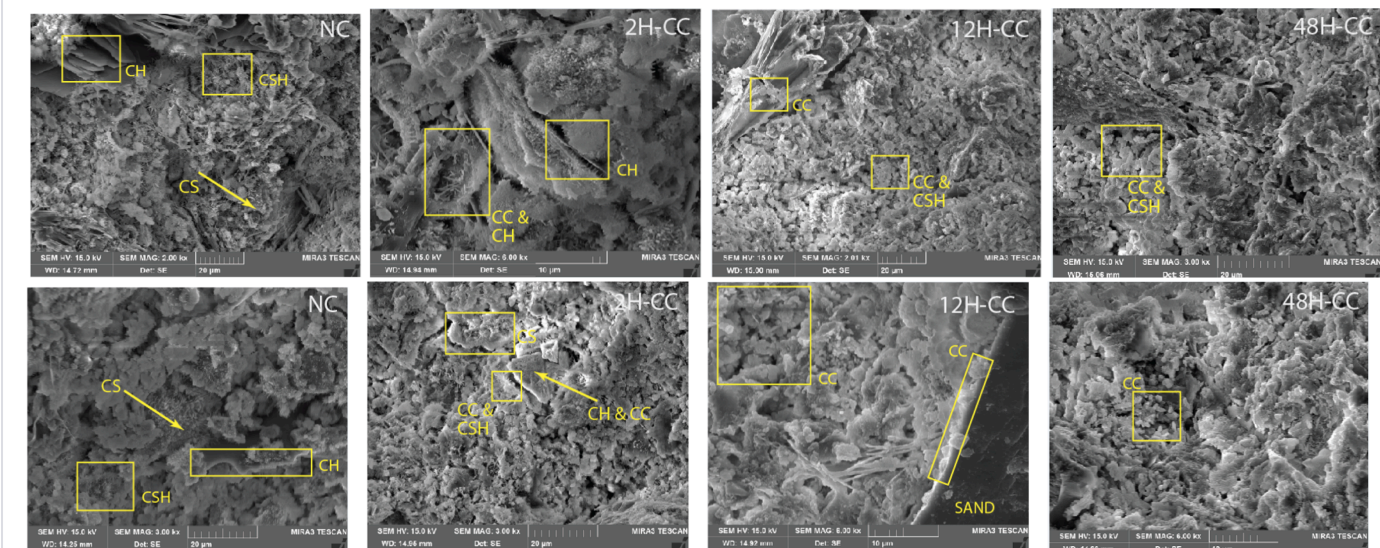
Slag absorbs more CO₂. Uptake (bars) climbs fastest in the first 12 hours, and the 20% slag mix leads at every duration.

FIGURE 3 · CARBON AND COST



Lower carbon, cost offset. Net CO₂ falls by about 161, 248 and 320 kg per tonne for 0%, 10% and 20% slag. Added cost drops from \$18.08 to \$5.94 per tonne as slag replaces clinker.

FIGURE 4 · MICROSTRUCTURE OF THE 20% SLAG MIX (SEM, 28 DAYS)



Carbonates fill the gaps. Left to right: normal curing, then 2, 12 and 48 hours of CO₂ curing. Calcium hydroxide flakes give way to calcium carbonate (CC), and fibrous C-S-H is replaced by a denser intergrown matrix as curing time increases.